

Drawing Tools

LINE • CIRCLE • ARC • PLINE • HATCH

Coordinates | Drawing Methods | Precision Input | Keyboard Shortcuts

What You Will Learn

- All 8 basic drawing commands: LINE, CIRCLE, ARC, RECTANG, PLINE, POLYGON, ELLIPSE, HATCH
- Three coordinate input methods: absolute (X,Y), relative (@X,Y), and polar (@dist<angle)
- Five CIRCLE creation methods and key ARC methods with their command syntax
- How to use PLINE for connected segments and HATCH to fill enclosed areas with patterns

Key Keyboard Shortcuts — Drawing Commands

L	= LINE	C	= CIRCLE	A	= ARC	REC	= RECTANG
PL	= PLINE	POL	= POLYGON	EL	= ELLIPSE	H	= HATCH

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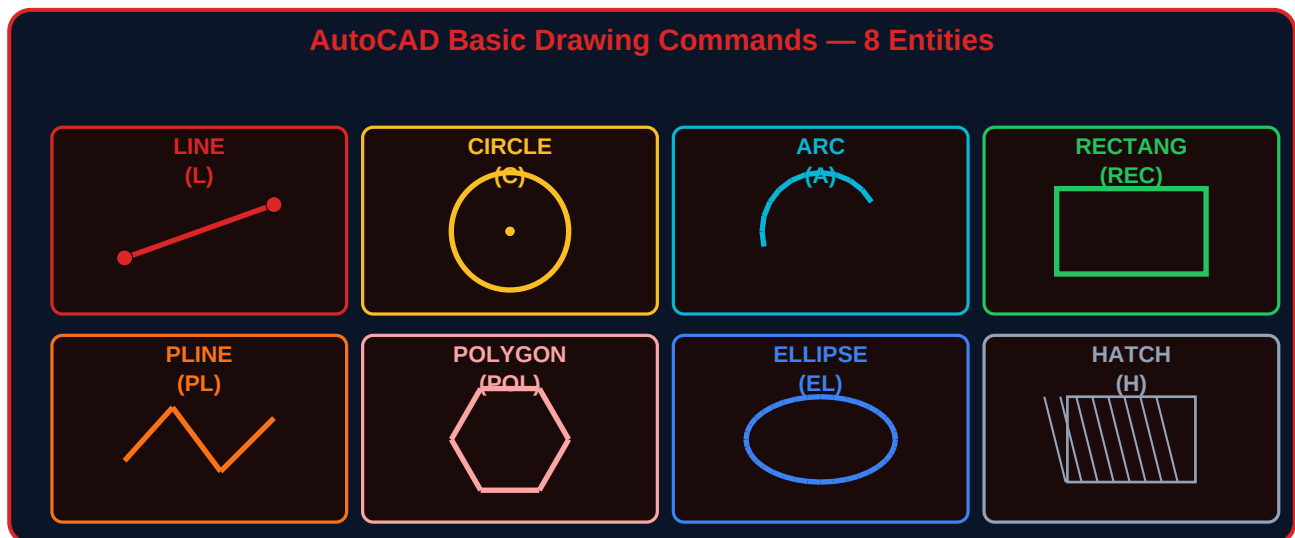
CAD-003: Drawing Tools in AutoCAD

AutoCAD's drawing commands are the foundation of every design. Everything you create — walls, pipes, structural members, equipment outlines — begins as a basic drawing entity: a line, circle, arc, or rectangle. In this lesson you will master all eight fundamental drawing commands and the three coordinate input methods that give you precise control over where every object is placed. These skills are used on every AutoCAD drawing you will ever produce.

Learning Objectives

- Use the LINE, CIRCLE, ARC, and RECTANG commands with all available creation options
- Enter coordinates using absolute (X,Y), relative (@X,Y), and polar (@dist<angle) methods
- Create continuous paths with PLINE, regular shapes with POLYGON, and curved shapes with ELLIPSE
- Apply HATCH patterns to filled enclosed areas for materials, sections, and site coverage
- Use Direct Distance Entry (DDE) and ortho mode (F8) for fast accurate drawing

1. The Eight Basic Drawing Commands

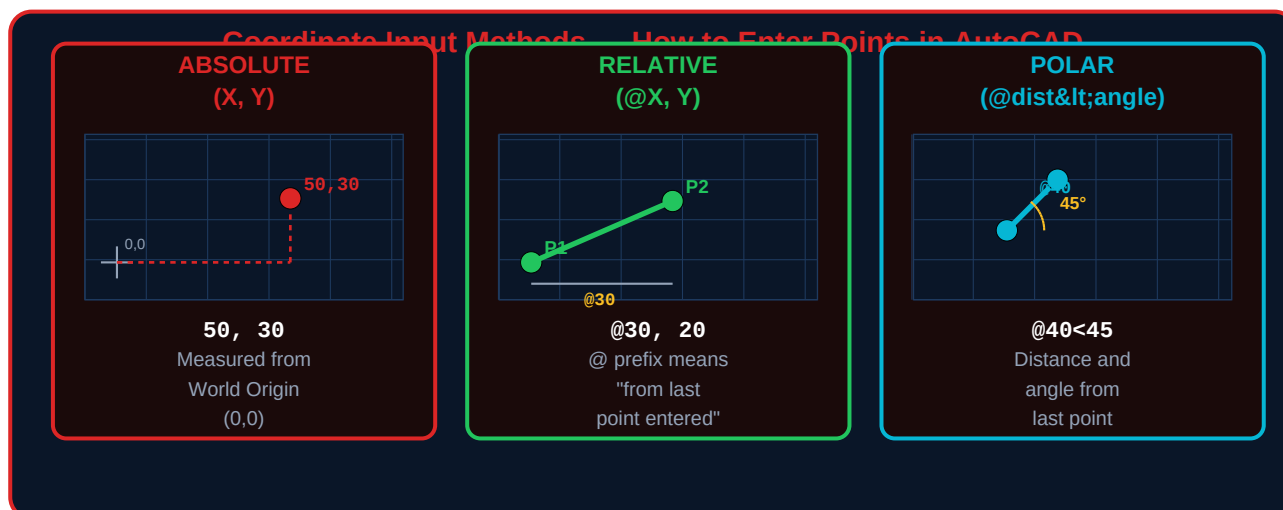


Command	Alias	What It Creates	Key Option
LINE	L	Individual straight line segments. Each segment is a separate object.	Close (C): closes a series of lines back to the first point.
CIRCLE	C	A complete circle. Six creation methods (see next page).	Diameter (D): switch from radius to diameter input after placing centre.
ARC	A	A circular arc (partial circle). Many creation methods.	Default: 3-point arc. Counterclockwise by default.

RECTANG	REC	A closed rectangle drawn as a single polyline object.	Area (A): specify by area + one side length. Fillet (F): rounded corners.
PLINE	PL	A polyline: a connected series of lines and arcs treated as one object.	Arc (A): switch to arc mode mid-command to draw curved segments.
POLYGON	POL	A regular polygon with equal sides (3 to 1024 sides).	Inscribed (I): fits inside a circle. Circumscribed (C): fits outside.
ELLIPSE	EL	An ellipse or elliptical arc.	Arc (A): draw an elliptical arc instead of a full ellipse.
HATCH	H	Fills an enclosed area with a repeating pattern or solid fill.	Pattern type: ANSI31 (steel), ANSI37 (insulation), SOLID (solid fill), EARTH.

2. Coordinate Input — How to Locate Points Precisely

Precise drawing in AutoCAD requires telling the software exactly where to place each point. There are three coordinate input methods. All three can be mixed freely within a single command — choose the method that is most convenient for the current step.



Method	Syntax	Example	When to Use
Absolute coordinates	X,Y	50,30	When you know the exact global position. Rare in practice — most drawings are drawn relative to existing geometry.
Relative coordinates	@X,Y	@30,20	Most common method. Moves from the last point by a specified X and Y offset. @0,50 moves straight up 50 units.
Relative polar	@dist<angle	@40<45	When you know the length and angle of a line. @100<0 = horizontal 100 units right. @100<90 = vertical 100 units up.
Direct Distance Entry	Type distance + Enter	Point mouse right, type 100, Enter	Fastest method for horizontal/vertical lines. Move the mouse in the direction you want, type the distance, press Enter. Requires Ortho (F8) for strict H/V.

3. Ortho Mode, Polar Tracking, and Object Snap — Precision Aids

Tool	Toggle	What It Does	When to Use
Ortho Mode	F8	Restricts cursor movement to exactly horizontal or vertical. Cursor can only move 0°, 90°, 180°, or 270°.	Drawing rectangular geometry — walls, rooms, floor plans. Fastest mode for orthogonal drawings.

Polar Tracking	F10	Snaps cursor to user-defined angles (e.g. 15°, 30°, 45°, 90°). Shows dashed line and angle tooltip.	Drawing diagonal lines at standard angles without typing polar coordinates each time.
Object Snap (OSNAP)	F3	Snaps cursor to exact geometric points on existing objects: endpoints, midpoints, centres, intersections, perpendicular, tangent.	All professional drawing. NEVER turn off OSNAP when connecting to existing geometry.
Dynamic Input	F12	Shows coordinate and angle input fields directly at the cursor on screen, rather than in the command line.	Preferred by many users — avoids looking down at the command line. Same coordinate entry, different display location.

4. LINE Command — Step by Step

The LINE command creates individual straight line segments. Each segment is drawn from point to point. You can draw multiple connected segments in one command — press Enter to end the command or type **C** to close back to the first point.

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Command: L → Enter
Specify first point: 0,0 → Enter
Specify next point or [Undo]: @100,0 → Enter (horizontal 100mm right)
Specify next point or [Undo]: @0,60 → Enter (vertical 60mm up)
Specify next point or [Close/Undo]: C → Enter (close back to 0,0)
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5. CIRCLE and ARC — Creation Methods

CIRCLE and ARC Creation Methods

CIRCLE — 5 Methods

1

Centre, Radius
Type C → click centre → type 50

2

Centre, Diameter
Type C → click → type D then 100

3

3 Points (3P)
Type C 3P → click 3 points on circumference

4

2 Points (2P)
Type C 2P → click 2 diameter endpoints

5

TTR (Tan,Tan,R)
Type C TTR → select 2 tangent objects → radius

ARC — Key Methods

 **3 Points (default)**
Type A → start → midpoint → end point

 **Start, Centre, End**
Type A S → start → C → centre → end

 **Start, Centre, Angle**
Type A S → start → C → centre → angle (e.g. 90)

 **Start, End, Radius**
Type A S → start → E → end → R → radius

 **Continue (from last)**
Press A → Enter without points (tangent continuation)

6. RECTANG Command

Option	Syntax During Command	Use Case
Corner to corner	Click P1 (first corner) → type @width,height	Fastest for simple rectangles: REC → 0,0 → @300,200 draws a 300×200mm rectangle from origin.
With fillet	REC → F → enter radius → draw rectangle	Rounded corners for architectural features, furniture symbols, rounded equipment outlines.
With chamfer	REC → CHA → enter distances → draw rectangle	Chamfered (cut diagonal corner) for certain structural details and mechanical profiles.
Specify area	REC → A → type area → type L (length) → enter length	When you know the total area and one dimension: draw a room of exactly 20m².

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Page 6

Specify rotation	REC → R → enter angle → draw	Rotate the rectangle to a non-standard angle (e.g. buildings not aligned to grid).
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7. PLINE (Polyline) — Connected Segments as One Object

A polyline is a connected sequence of straight and/or arc segments that AutoCAD treats as a single object. This is important: selecting any part of the polyline selects the whole thing. Polylines can have width (useful for walls in plans). All closed shapes drawn with RECTANG or POLYGON are actually polylines.

PLINE Sub-command	When It Appears	What It Does
Arc (A)	During PLINE while drawing straight segments	Switch to arc mode — draw a tangent arc from the last point. Type L to return to straight line mode.
Close (C)	After at least 2 segments have been drawn	Closes the polyline back to the first point with a final segment.
Width (W)	During PLINE command	Sets the start and end width of the next segment. Useful for site plans (wide roads), schematic arrows, and graphical elements.
Halfwidth (H)	During PLINE command	Sets half the total width. Easier than calculating full width when you know the half-width.
Undo (U)	During PLINE command	Undoes the last segment drawn without ending the command. Allows correction without starting over.

8. POLYGON, ELLIPSE, and HATCH

Command	Step-by-Step	Engineering Application
POLYGON	1. Type POL → Enter 2. Enter number of sides (e.g. 6) 3. Click or type centre point 4. Choose I (Inscribed) or C (Circumscribed) 5. Type the radius → Enter	Bolt heads (hexagonal), nut profiles on mechanical drawings, column cross-sections, geometric site features, logo elements.
ELLIPSE	1. Type EL → Enter 2. Click or type centre point 3. Type one axis endpoint distance 4. Type the other axis endpoint distance	Oval columns, elliptical arches, planting symbols on landscape plans, isometric circles, drain/manhole symbols in plan.

HATCH	1. Type H → Enter 2. Opens Hatch and Gradient dialog or ribbon panel 3. Select pattern: ANSI31 (steel lines), SOLID, EARTH, etc. 4. Set scale (controls pattern density) 5. Click "Add: Pick Points" → click inside closed area → Enter	Section views (hatching shows cut material), floor finish indication, landscaping areas, site coverage, insulation indication in details.
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Common HATCH Patterns Reference

Pattern Name	Represents	Typical Scale	Application
ANSI31	General metal/steel (lines at 45°)	1–5	Steel sections, metal components in sections
ANSI37	Insulation (irregular wavy lines)	1–5	Thermal insulation in wall/roof sections
SOLID	Solid fill (black or colour)	N/A	Filled shapes, legends, walls in plans
EARTH	Earth/soil (triangles and dots)	1–10	Ground conditions in site sections, earthworks
CONCRETE	Concrete aggregate pattern	1–10	Concrete sections, foundations, slabs
BRASS	Brass (parallel lines)	1–5	Brass and non-ferrous metal components
STARS	Star pattern	5–20	Floor tile patterns, decorative elements

9. Practice Exercises

#	Exercise	Step-by-Step Instructions
1	Draw a rectangle using three methods	(a) Use REC → 0,0 → @200,150 to draw a 200×150mm rectangle from the origin. (b) Use LINE and relative coordinates to draw the same rectangle. (c) Use RECTANG with Fillet=20 to draw a rectangle with rounded corners. Compare: the RECTANG result is one object; the LINE result is four objects.
2	Circle and arc practice	(a) Draw a circle using Centre,Radius method: centre at 100,100, radius 50. (b) Draw a circle using 3-Point method through three click points. (c) Draw a 90° arc using Start, Centre, Angle method. (d) Draw a full floor plan door symbol: a straight line for the door + a quarter-circle arc for the swing.
3	Coordinate input methods	Draw a regular hexagonal bolt head outline: (a) Type POL → 6 → 0,0 → I → 25 (inscribed, radius 25mm). (b) Draw a line from the centre to one vertex using polar coordinates (@25<0). (c) Draw a circle at 0,0 with radius 12 (representing the bolt shank).
4	Polyline with arc	Draw a furniture symbol for a sofa: use PLINE to draw a U-shape. Start at 0,0 → @0,120 → @80,0 → @0,-120 (three sides of rectangle). Then switch to Arc mode (A) → draw a semicircle back to 0,0. The result should be a closed sofa outline as a single polyline.
5	Hatch a floor plan room	(a) Draw a closed rectangle (4000×5000mm) representing a room. (b) Apply HATCH with pattern AR-HBONE (herringbone) for a timber floor effect. (c) Apply HATCH SOLID to represent a wall (draw the wall as a closed rectangle first). (d) Adjust hatch scale to make the pattern look correctly proportioned.

Lesson Summary

Command / Method	Key Rule to Remember
LINE (L)	Each segment is a separate object. Use C to close. Use @X,Y for most inputs.
CIRCLE (C)	5 methods. Most common: Centre+Radius. Type D to switch to diameter input after centre.
ARC (A)	Default = 3-point arc. Many other methods available. Draws counterclockwise by default.
RECTANG (REC)	Draws a rectangle as a single closed polyline. Use @width,height for second corner.
PLINE (PL)	Creates a connected path as ONE object. Type A mid-command for arc segments.
POLYGON (POL)	Enter number of sides → centre → I or C → radius. 3 sides = triangle, 6 = hexagon.
ELLIPSE (EL)	Enter centre → major axis endpoint distance → minor axis endpoint distance.

HATCH (H)	Area must be fully enclosed before hatching. Set scale to match drawing scale.
Absolute coords	X,Y — measured from World Origin (0,0). Use for the FIRST point of a new object.
Relative coords	@X,Y — offset from LAST point. Use for ALL subsequent points.
Polar coords	@dist<angle — distance and angle from last point. @100<90 = 100 units straight up.

What Is Next — CAD-004

CAD-004: Modify Tools — The essential editing commands every AutoCAD user needs: ERASE, MOVE, COPY, OFFSET, MIRROR, ROTATE, SCALE, TRIM, EXTEND, STRETCH, FILLET, CHAMFER, ARRAY (rectangular and polar). Grip editing for fast direct manipulation without commands.

Resource	Details
AutoCAD PDF Series	CAD-001 through CAD-030 — ayetechub.com/pdfs.html
Free AutoCAD	AutoCAD Web (browser-based) — autodesk.com/products/autocad/web-app
Community	Telegram: t.me/ayetechub